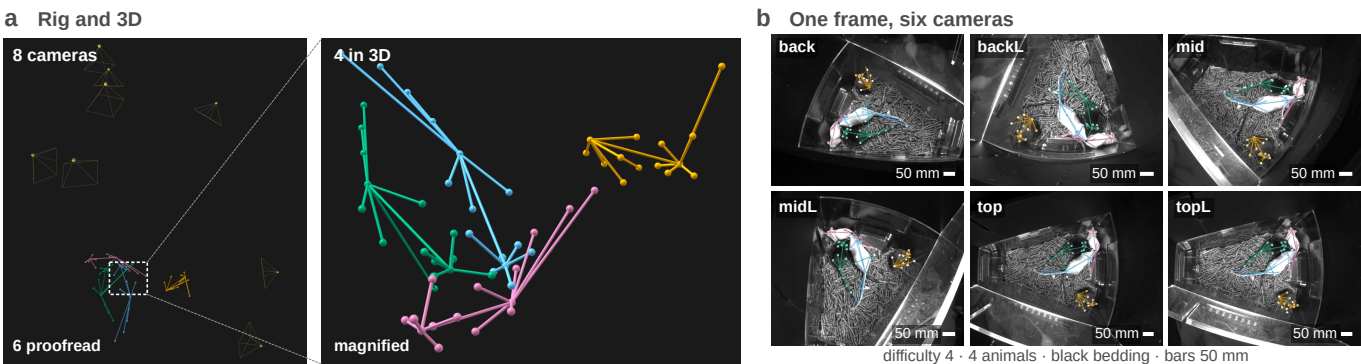
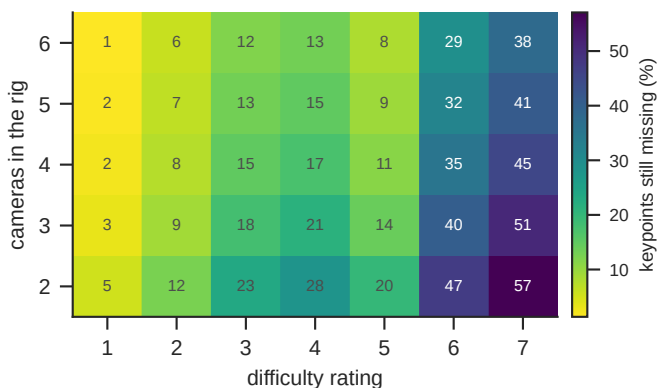
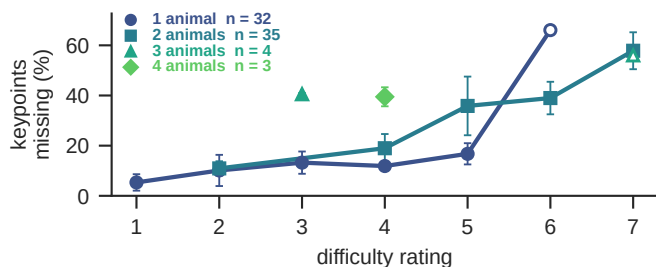




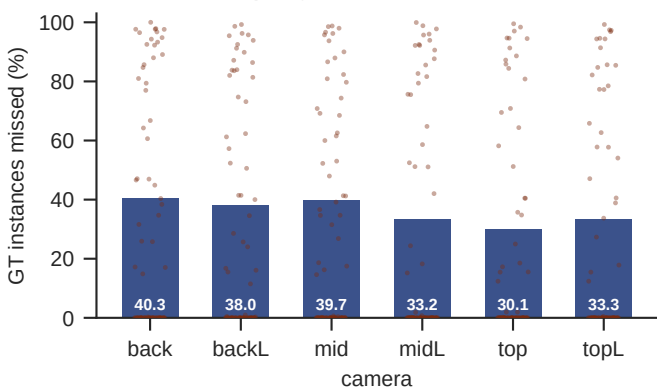
**c Missing keypoints vs difficulty and cameras**



**d Animal-count control**



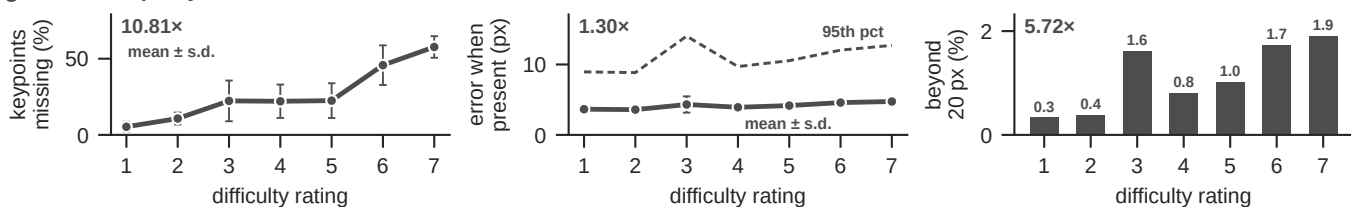
**e Per-camera detection quality**



**f Corpora**

|                  | BMimica       | SLAP-2M         | Total      |
|------------------|---------------|-----------------|------------|
| Cameras          | 5             | 8 (6 proofread) | —          |
| Animals          | 2             | 1–4             | 1–4        |
| Sessions with 3D | 56 of 56      | 74 of 74        | 130 of 130 |
| Frames           | 10,084,734    | 1,954,440       | 12,039,174 |
| Hours            | 18.7 @150 fps | 10.9 @50 fps    | 29.6       |
| Nodes            | 15            | 15              | 15         |

**g Detection quality**



±1 s.d. between sessions · n = 12, 13, 9, 13, 10, 4, 13 sessions at difficulty 1–7

**h Difficulty strata**

| Difficulty | Sessions | Keypoints  | Animals | Bedding (black/white) | Missing (%) | Error mean (px) | Error p95 (px) | Error p99 (px) | > 20 px (%) |
|------------|----------|------------|---------|-----------------------|-------------|-----------------|----------------|----------------|-------------|
| 1          | 12       | 30,199,536 | 1       | 12/0                  | 5.3         | 3.65            | 8.95           | 14.45          | 0.33        |
| 2          | 13       | 40,786,166 | 1, 2    | 8/5                   | 10.7        | 3.59            | 8.84           | 14.58          | 0.38        |
| 3          | 9        | 27,668,941 | 1, 3    | 4/5                   | 22.3        | 4.32            | 14.02          | 22.49          | 1.61        |
| 4          | 13       | 29,355,207 | 1, 2, 4 | 6/7                   | 22.1        | 3.92            | 9.70           | 18.38          | 0.80        |
| 5          | 10       | 32,791,049 | 1, 2    | 6/4                   | 22.5        | 4.16            | 10.54          | 20.48          | 1.02        |
| 6          | 4        | 11,091,948 | 1, 2    | 1/3                   | 45.8        | 4.59            | 12.04          | 26.42          | 1.72        |
| 7          | 13       | 15,241,535 | 2, 3    | 7/6                   | 57.7        | 4.74            | 12.71          | 27.77          | 1.90        |